

Research Memo: In-Context Learning and Query Vector Movement

Theoretical Synthesis and Experimental Design

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Abstract

This document captures a comprehensive research discussion bridging modern In-Context Learning (ICL) in Large Language Models with classical Query Vector Movement (QVM) theories. It includes a literature review of recent ICL advancements, a structured reading guide, a detailed breakdown of ICL mechanics, and a formal synthesis connecting attention dynamics to relevance feedback, concluding with proposed ablation studies.

1 Recent Literature on In-Context Learning (ICL)

Query: Give me the most recent papers on In Context Learning (ICL).

Here are some of the most significant recent papers on In-Context Learning (ICL), highlighting new methodologies, theoretical frameworks, and applications in multi-modal architectures:

- **Many-Shot CoT-ICL: Making In-Context Learning Truly Learn (Chung, n.d.):** Investigates ICL behavior when scaled to dozens or hundreds of examples. Proposes Curvilinear Demonstration Selection (CDS) to restructure the context window into a learning curriculum for chain-of-thought reasoning.
- **Parallel In-context Learning for Large Vision Language Models (Yamaguchi, n.d.):** Tackles the quadratic computational cost of multi-modal ICL. The Parallel-ICL algorithm partitions contexts, processes chunks in parallel, and merges predictions using a weighted Product-of-Experts ensemble.
- **On Theoretical Interpretations of Concept-Based In-Context Learning (Tang et al., 2025):** Introduces a mathematical framework quantifying how models leverage prompt knowledge and proposes a similarity measure between demonstrations and queries.

Foundational context is provided by Kossen et al. (2023) on label relationships, Lin & Lee (2024) on dual operating modes, and Dong et al. (2022) for a comprehensive general survey.

2 Recommended Reading Progression

Query: Give me a rough order of reading since I want to learn about the topic and then look at recent advancements.

To understand ICL from the ground up, the following progression is recommended:

1. Core Foundations: The Birth of ICL

- *Language Models are Few-Shot Learners (Brown et al., 2020)*: The seminal GPT-3 paper introducing ICL.
- *It's Not Just Size That Matters: Small Language Models Are Also Few-Shot Learners (Schick & Schütze, 2021)*: Introduces Pattern-Exploiting Training (PET) for smaller models.

2. Surveys & Syntheses: Mapping the Landscape

- *A Survey on In-context Learning (Dong et al., 2022)*: The definitive early survey formalizing ICL definitions and prompt design.
- *A Practical Survey on Zero-shot Prompt Design for In-context Learning (Li, 2023)*: Focuses on discrete and continuous prompts.

3. Advanced Mechanics: How Does it Actually Work?

- *Rethinking the Role of Demonstrations (Min et al., 2022)*: Argues that demonstrations primarily teach format and label space, not the input-output mapping.
- *In-Context Learning Learns Label Relationships but Is Not Conventional Learning (Kossen et al., 2023)*: Provides a counter-argument that ICL does rely on label information in certain scenarios.

4. The Current Frontier: Many-Shot & Multi-Modal (2024–2026)

- *A Survey on Multimodal Large Language Models (Yin et al., 2023)*: Covers the transition to Multimodal ICL (MM-ICL).
- *Many-Shot In-Context Learning (Agarwal et al., 2024)*: Explores scaling to thousands of demonstrations.
- *From Few to Many: Self-Improving Many-Shot Reasoners (Wan et al., 2025)*: Explores automated prompt construction and demonstration optimization.

3 Deep Dive: A Survey on In-context Learning (Dong et al., 2022)

Query: Explain this paper to me section by section in detail providing intuition and math married together beautifully...

1. Introduction & Formulation

Intuition: ICL allows models to learn without parameter updates, functioning similarly to an amnesiac student recognizing patterns from a few solved examples on a worksheet.

Math: In ICL, parameters θ are frozen. A context $C = \{(x_1, y_1), \dots, (x_k, y_k)\}$ is concatenated with a target query x_{target} . The objective is to compute:

$$y^* = \arg \max_y P(y|x_{target}, C; \theta) \quad (1)$$

2. Prompt Design

The exact composition of C is paramount, divided into three sub-problems:

- **Demonstration Selection:** Finding examples conceptually closest to the target query using unsupervised distance metrics (e.g., L2, cosine similarity) or supervised retrieval models.
- **Demonstration Formatting:** Mapping inputs and labels into a natural language template (discrete prompts) or vector space (continuous prompts).
- **Demonstration Ordering:** Mitigating recency bias, where models place disproportionate weight on the final example preceding the query.

3. Inference & Scoring Strategies

Math: Contrasting direct scoring with channel scoring to mitigate prior probability bias:

$$\text{Direct Scoring: } Score = P(y|x, C) \quad (2)$$

$$\text{Channel Scoring: } Score = P(x|y, C) \cdot P(y|C) \quad (3)$$

Channel scoring naturally normalizes output distribution and mitigates bias toward common words.

4. Pre-training for ICL

ICL capabilities scale with pre-training corpora containing long-range dependencies and implicit analogies. Meta-tuning on ICL-formatted data can further sharpen pattern recognition.

5. Under the Hood: Implicit Bayesian Inference and Attention

Math: The context C helps infer the posterior distribution of a latent task concept γ :

$$P(y|x_{target}, C) \approx \int P(y|x_{target}, \gamma)P(\gamma|C)d\gamma \quad (4)$$

Furthermore, the self-attention mechanism strongly parallels standard gradient descent, computing a pseudo-gradient to adapt representations dynamically.

4 Theoretical Synthesis: Formalizing ICL via QVM

Query: So icl can effectively be seen as an implicit movement of embeddings in the latent space. I want to delve deeper into this. There is a theory called Query Vector Movement... TPAMI 2005 paper by Bir Bhanu.

Drawing a direct mathematical bridge between classic Content-Based Image Retrieval (CBIR) Relevance Feedback mechanisms (QVM) and modern LLM attention dynamics.

Formalizing ICL as Query Vector Movement

In classical QVM (e.g., Rocchio algorithm), an initial query vector q_{old} is updated based on positive (D^+) and negative (D^-) relevance feedback:

$$q_{new} = \alpha q_{old} + \beta \frac{1}{|D^+|} \sum_{x \in D^+} x - \gamma \frac{1}{|D^-|} \sum_{x \in D^-} x \quad (5)$$

In a Transformer layer, for a target query token representation h_q , the residual connection and attention mechanism update is:

$$h_q^{(new)} = h_q^{(old)} + \sum_{i \in C} \text{softmax} \left(\frac{(W_q h_q^{(old)})(W_k h_i)^T}{\sqrt{d}} \right) W_v h_i \quad (6)$$

The Mathematical Symmetry:

- **The Residual is q_{old} :** $h_q^{(old)}$ is analogous to αq_{old} (where $\alpha = 1$).
- **Attention as Relevance Weighting:** The summation over context tokens C represents the query vector moving toward “positive” demonstrations. The static weight $\beta \frac{1}{|D^+|}$ is replaced by the dynamic attention score $\text{softmax}(\dots)$.
- **The Missing γ Term:** Standard ICL lacks the $-\gamma$ term because negative demonstrations are rarely provided. Engineering negative context representations to explicitly repulse the query vector presents a significant research gap.

Probing the Sensitivity: Experimental Design

To formally demonstrate ICL sensitivity through the lens of QVM, three ablation studies are proposed:

1. **Latent Noise Injection (Continuous Perturbation):** Inject scaled Gaussian noise into the value vectors $V_i = W_v h_i$ of the context. Track the cosine similarity between the clean $h_q^{(new)}$ and perturbed $\hat{h}_q^{(new)}$ across layers.
2. **Attention Distribution Skewing (Structural Perturbation):** Permute the order of demonstrations in C . Measure the variance of the final query vector’s position to mathematically quantify recency bias.
3. **Label Space Flipping (Semantic Perturbation):** Replace correct labels with incorrect ones. Project intermediate query embeddings into a 2D space to observe if the vector moves toward the format manifold or is pulled toward the incorrect semantic manifold.